

MATH 324 Winter 2025 - Tutorial 3

Section 003 (Thursday)

January 30, 2025

1 Confidence Intervals

1.1 Probability integral transform

Suppose X is a continuous random variable with CDF F . The *probability integral transform* is the result that

$$F(X) \sim U(0, 1).$$

Thus, $F(X)$ is itself a pivotal quantity which depends on the CDF of X but whose distribution contains none of X 's (unknown) parameters. With this pivotal quantity, we can then generate confidence intervals for the parameters. As seen in class, $X \sim \text{Exp}(\lambda)$, $2n\lambda\bar{X}_n \sim \chi_{2n}^2$ allows us to construct confidence intervals for λ in the parametrization where $\mathbb{E}[X] = 1/\lambda$.

An aside... Why do we use the probability integral transform? We've seen that it can easily provide a recipe for a pivotal quantity for a general continuous random variable. A common use case for the PIT is to generate samples from a continuous random variable: software can easily generate $U(0, 1)$ observations. Using these, we can apply the inverse of the CDF (if available!) to get samples from the desired continuous random variable.

Question 1 - Probability integral transform proof

Show that if X is a continuous random variable with CDF F , $F(X) \sim U(0, 1)$.

Solution

Let $Y = F(X)$. F is continuous and increasing on $[0, 1]$, as it is the CDF applied to a continuous random variable. Thus, it has an inverse on $[0, 1]$.

$$\begin{aligned} F_Y(y) &= \mathbb{P}(Y \leq y) \\ &= \mathbb{P}(F(X) \leq y) \\ &= \mathbb{P}(X \leq F^{-1}(y)) \\ &= F(F^{-1}(y)) \\ &= y \end{aligned}$$

We see this is the CDF of $U(0, 1)$ on $[0, 1]$. Meanwhile, $P(Y \leq y) = 0$ if $y \leq 0$ and $P(Y \geq y) = 1$ if $y \geq 1$. Thus, we recover the entire uniform CDF.

Question 2 - Distribution transformation

Suppose $X_1, \dots, X_n$ are i.i.d. continuous random variables with common CDF F . Show that

$$-2 \sum_{i=1}^n \ln F(X_i) \sim \chi_{2n}^2.$$

Solution

We know that $Y_i = F(X_i) \sim U(0, 1)$ for all i by the PIT result. Let $U_i = -2 \ln Y_i$. There are two approaches to finding the distribution of this transformation:

1. Method of transformations: If $U = h(Y)$, then

$$f_U(u) = f_Y(h^{-1}(u)) \left| \frac{dh^{-1}(u)}{du} \right|$$

In this case, we have that $h^{-1}(u) = e^{-u/2}$ and so $\frac{dh^{-1}(u)}{du} = \frac{-1}{2}e^{-u/2}$. Since Y is a uniform, its pdf is 1 between 0 and 1. Therefore,

$$f_U(u) = 1 \left| \frac{-1}{2}e^{-u/2} \right| = \frac{1}{2}e^{-u/2}$$

which is the pdf of an Exponential(1/2) random variable, or equivalently, the χ^2 with 2 degrees of freedom.

2. Method of CDF: We directly compute

$$\mathbb{P}(U \leq u) = \mathbb{P}(-2 \ln Y \leq u) = \mathbb{P}(Y > e^{-u/2}) = 1 - e^{-u/2},$$

as $\mathbb{P}(Y \geq y) = 1 - y$ for $Y \sim U(0, 1)$. This is the CDF of the Exponential(1/2), which is also the chi-squared with 2 degrees of freedom.

Now that we've established $U_i \sim \chi_2^2$ for all i , we use the property that if $X_i \sim \chi_{k_i}^2$ then $\sum_{i=1}^n X_i \sim \chi_{k_1 + \dots + k_n}^2$. This means that

$$\sum_{i=1}^n U_i \sim \chi_{2+\dots+2}^2 = \chi_{2n}^2.$$

1.2 Confidence interval for σ^2

By the nature of the variance, we would want a strictly positive confidence interval for σ^2 . Theorem 7.3 tells us

$$\frac{\sum_{i=1}^n (Y_i - \bar{Y})^2}{\sigma^2} = \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$$

The confidence interval is constructed so that it is the shortest interval containing σ^2 with probability $1 - \alpha$ (as the χ^2 distribution is not symmetric). We have

$$\mathbb{P}\left(\chi_{1-\alpha/2}^2 \leq \frac{(n-1)S^2}{\sigma^2} \leq \chi_{\alpha/2}^2\right) = 1 - \alpha/2.$$

Thus, our 2-sided C.I. for σ^2 is

$$\left(\frac{(n-1)S^2}{\chi_{\alpha/2}^2}, \frac{(n-1)S^2}{\chi_{1-\alpha/2}^2}\right).$$

Question 3 - Choosing a sample size for difference in proportions

(Exercise 8.72) Telephone pollsters often interview between 1000 and 1500 individuals regarding their opinions on various issues. Does the performance of colleges' athletic teams have a positive impact on the public's perception of the prestige of the institutions? A new survey is to be undertaken to see if there is a difference between the opinions of men and women on this issue.

- (a) If 1000 men and 1000 women are to be interviewed, how accurately could you estimate the difference in the proportions who think that the performance of their athletics teams has a positive impact on the perceived prestige of the institutions? Find a bound on the error of estimation.
- (b) Suppose that you were designing the survey and wished to estimate the difference in a pair of proportions, correct to within 0.02, with probability 0.9. How many interviewees should be included in each sample?

Solution

- (a) Since the true proportions are unknown, we will use $p_1 = p_2 = 0.5$, as this maximizes the terms $p(1-p)$ in the joint variance (this will also let us have the largest possible error bound). We can choose either a 2- σ error bound, or a 95% bound since n is larger. Using the 95% confidence interval, $z_{\alpha/2} = 1.96$ and our bound is

$$1.96\sqrt{\frac{0.5(0.5)}{1000} + \frac{0.5(0.5)}{1000}} = 0.044.$$

- (b) We again use $p_1 = p_2 = 0.5$ and will assume that $n_1 = n_2 = n$. For the 90% probability, $z_{\alpha/2} = 1.645$. Then we solve for n in

$$\begin{aligned} 1.645\sqrt{\frac{0.5(0.5)}{n} + \frac{0.5(0.5)}{n}} &= 0.02 \\ \frac{2(0.5)^2}{n} &= \left(\frac{0.02}{1.645}\right)^2 \\ n &= 3383 \end{aligned}$$

Question 4 - Confidence interval for σ^2

(Exercise 8.103) A precision instrument is guaranteed to read accurately to within 2 units. A sample of four instrument readings on the same object yielded the measurements 353, 351, 351, and 355. Find a 90% confidence interval for the population variance. What assumptions are necessary? Does the guarantee seem reasonable?

Solution

We can calculate the sample variance

$$s^2 = \frac{1}{4-1} \left(\sum_{i=1}^4 y_i^2 - n(\bar{y})^2 \right) = 3.67.$$

We also have that $\chi_{0.95,3}^2 = 0.3518$ and $\chi_{0.05,3}^2 = 7.815$. Therefore, our 90% confidence interval is

$$\left(\frac{3(3.67)}{7.815}, \frac{3(3.67)}{0.35185} \right) = (1.4, 31.3).$$

We assume the measurements have a normal distribution and are independent. The guarantee does not seem reasonable as the interval for the standard deviation would be about (1.2,5.6). If a standard deviation of over 5 is still likely, then we cannot guarantee accuracy to 2 units.

2 Properties of Point Estimators

2.1 Relative efficiency

Definition: Given $\hat{\theta}_1$ and $\hat{\theta}_2$ unbiased for θ , the *efficiency* of $\hat{\theta}_1$ relative to $\hat{\theta}_2$ is

$$\text{eff}(\hat{\theta}_1, \hat{\theta}_2) = \frac{\text{Var}(\hat{\theta}_2)}{\text{Var}(\hat{\theta}_1)}$$

Note that the **first** term is the **denominator**!

If $\text{eff}(\hat{\theta}_1, \hat{\theta}_2) > 1$, then $\hat{\theta}_1$ is the better estimator. ($\text{Var}(\hat{\theta}_2) > \text{Var}(\hat{\theta}_1)$ and both are unbiased, so MSE is lower for $\hat{\theta}_1$.)

2.2 Consistency

Definition: $\hat{\theta}_n$ is a *consistent* estimator of θ if $\hat{\theta}_n \xrightarrow{p} \theta$. i.e.

$$\forall \varepsilon > 0, \quad \lim_{n \rightarrow \infty} \mathbb{P}(|\hat{\theta}_n - \theta| > \varepsilon) = 0$$

or

$$\lim_{n \rightarrow \infty} \mathbb{P}(|\hat{\theta}_n - \theta| \leq \varepsilon) = 1.$$

Theorem 9.1 If $\hat{\theta}$ is unbiased for θ and $\lim_{n \rightarrow \infty} \text{Var}(\hat{\theta}_n) = 0$, then $\hat{\theta}_n \xrightarrow{p} \theta$.

An important property for convergence in probability (i.e. $X_n \xrightarrow{p} X$) is linearity: if $X_n \xrightarrow{p} X$ and $Y_n \xrightarrow{p} Y$, $a, b \in \mathbb{R}$, then

$$aX_n + bY_n \xrightarrow{p} aX + bY.$$

Question 5 - Efficiency

(Exercise 9.2) Let $Y_1, Y_2, \dots, Y_n$ denote a random sample from a population with mean μ and variance σ^2 . Consider the following three estimators for μ :

$$\hat{\mu}_1 = \frac{1}{2}(Y_1 + Y_2) \quad \hat{\mu}_2 = \frac{1}{4}Y_1 + \frac{Y_2 + \dots + Y_{n-1}}{2(n-2)} + \frac{1}{4}Y_n, \quad \hat{\mu}_3 = \bar{Y}$$

- Show that each of the three estimators is unbiased.
- Find the efficiency of $\hat{\mu}_3$ relative to $\hat{\mu}_1$ and $\hat{\mu}_2$, respectively.

Solution

- This follows rather easily from linearity of the expectation.
- Note that independence is crucial to split the variance of sums!

$$\text{Var}(\hat{\mu}_1) = \text{Var}\left(\frac{1}{2}(Y_1 + Y_2)\right) = \frac{1}{4}(\text{Var}(Y_1) + \text{Var}(Y_2)) = \frac{\sigma^2}{2}$$

$$\text{Var}(\hat{\mu}_2) = \frac{1}{16}\text{Var}(Y_1) + \frac{1}{4}\frac{1}{(n-2)^2}[\text{Var}(Y_2) + \dots + \text{Var}(Y_{n-1})] + \frac{1}{16}\text{Var}(Y_n) = \frac{\sigma^2}{8} + \frac{\sigma^2}{4(n-2)}$$

$$\text{Var}(\hat{\mu}_3) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(Y_i) = \frac{\sigma^2}{n}$$

We calculate the relative efficiencies:

$$\text{eff}(\hat{\mu}_3, \hat{\mu}_1) = \frac{\text{Var}(\hat{\mu}_1)}{\text{Var}(\hat{\mu}_3)} = \frac{\sigma^2/2}{\sigma^2/n} = \frac{n}{2}$$

which is greater than 1 for all $n > 2$.

$$\text{eff}(\hat{\mu}_3, \hat{\mu}_2) = \frac{\text{Var}(\hat{\mu}_2)}{\text{Var}(\hat{\mu}_3)} = \frac{\sigma^2/8 + \sigma^2/(4(n-2))}{\sigma^2/n} = \frac{n^2}{8(n-2)}$$

which is greater than 1 when $n > 4$.

Question 6 - Consistency

(Exercise 9.17-9.18) Suppose that $X_1, X_2, \dots, X_n$ and $Y_1, Y_2, \dots, Y_n$ are independent random samples from populations with means μ_1 and μ_2 and variances σ_1^2 and σ_2^2 , respectively.

- (a) Show that $\bar{X} - \bar{Y}$ is a consistent estimator of $\mu_1 - \mu_2$.
- (b) Suppose that the populations are normally distributed with $\sigma_1^2 = \sigma_2^2 = \sigma^2$. Show that

$$\frac{\sum_{i=1}^n (X_i - \bar{X})^2 + \sum_{i=1}^n (Y_i - \bar{Y})^2}{2n - 2}$$

is a consistent estimator of σ^2 .

Solution

- (a) By the linearity of convergence in probability and weak law of large numbers, as $\bar{X}_n \xrightarrow{P} \mu_1$ and $\bar{Y}_n \xrightarrow{P} \mu_2$, then

$$\bar{X}_n - \bar{Y}_n \xrightarrow{P} \mu_1 - \mu_2.$$

- (b) We know that S^2 is unbiased for σ^2 . By theorem 7.3, $\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$. As the chi-squared distribution with ν degrees of freedom has variance 2ν , thus

$$\text{Var}(S^2) = \left(\frac{\sigma^2}{n-1} \right)^2 \text{Var} \left(\frac{n-1}{\sigma^2} S^2 \right) = \left(\frac{\sigma^2}{n-1} \right)^2 [2(n-1)] = \frac{\sigma^4}{n-1}$$

We check that the pooled variance

$$S_T^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2 + \sum_{i=1}^n (Y_i - \bar{Y})^2}{2n - 2}$$

is unbiased for σ^2 . Note that in this question, $n_1 = n_2 = n$.

$$\begin{aligned} \mathbb{E}[S_T^2] &= \mathbb{E} \left[\frac{\sum_{i=1}^n (X_i - \bar{X})^2 + \sum_{i=1}^n (Y_i - \bar{Y})^2}{2n - 2} \right] \\ &= \frac{1}{2n - 2} \mathbb{E} [(n-1)S_1^2 + (n-2)S_2^2] \\ &= \sigma^2 \end{aligned}$$

The variance is

$$\begin{aligned}
 \text{Var}(S_T^2) &= \frac{1}{(2n-2)^2} \text{Var} \left(\sum_{i=1}^n (X_i - \bar{X})^2 + \sum_{i=1}^n (Y_i - \bar{Y})^2 \right) \\
 &= \frac{1}{(2n-2)^2} \text{Var}((n-1)S_1^2) + \text{Var}((n-2)S_2^2) \quad \text{by independence} \\
 &= \frac{1}{(2n-2)^2} (n-1)^2 \text{Var}(S_1^2) + (n-2)^2 \text{Var}(S_2^2) \\
 &= \frac{\sigma^2}{2n-2}
 \end{aligned}$$

As S_T^2 is unbiased and its variance goes to 0 as $n \rightarrow \infty$, by theorem 9.3, S_T^2 is consistent for σ^2 .